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Episode 10 · September 23, 2026 · 5 papers

Mixed-integer flow formulations for motion planning and decision-making of networked multi-agent systems

SKIM

Angelo Caregnato-Neto et al.

The authors replace subtour-elimination constraints with network-flow constraints to keep a multi-vehicle MILP planner's communication graph connected, and find this scales better for plain connectivity but gives no average speedup for k-hop connectivity.

PROBLEM Centralized MILP planners for networked multi-agent systems must keep the inter-agent communication graph connected. The standard subtour elimination encoding needs a number of constraints that grows exponentially with fleet size, and existing k-hop encodings need binary variables that grow with the number of hops.

METHOD They add flow-based connectivity constraints to standard mixed-integer multi-vehicle maneuvering models used in receding horizon planning. One version handles standard connectivity and another handles k-hop connectivity, and they prove both encodings are necessary and sufficient, so every feasible network topology stays reachable. The key choice is to trade binary variables and exponential constraint families for many continuous flow variables, which yields polynomial constraint growth and a binary count that no longer depends on hop count.

RESULTS The abstract gives no numbers, only a statistical comparison with a commercial branch-and-bound solver on randomized environments with growing fleet sizes. The flow formulation beats subtour elimination for standard connectivity and solves larger fleets within the time limit, while the k-hop flow versions show no average improvement over the baseline despite a better theoretical worst case.

CAVEAT The main scaling claim rests on constraint counts and a worst-case iteration bound. The empirical side is one commercial solver on randomized synthetic environments, and the abstract does not report fleet sizes, time limits or optimality gaps. The k-hop result is essentially negative, and there appears to be no test of receding horizon behaviour under real communication loss or of decentralized solving.

WHY IT MATTERS This work is mostly peripheral to you. It treats the communication graph as a connectivity constraint inside one centralized optimizer, not as a channel for exchanging intents or negotiating maneuvers, and it says nothing about message design, loss or partial participation. It is only worth a look if you use MILP-based joint trajectory planning and need to add a connectivity requirement for platoons or convoys.

arXiv:2609.24474v1 PDF incremental

Hardware-in-the-Loop Evaluation of Game-Theoretic Autonomous Driving

SKIM

Aakanksha Kataria, Huiwen Yan, Mushuang Liu

The authors take existing Nash and Stackelberg intersection-crossing controllers, run them on Quanser QCar 2 scale cars through a staged pipeline from MATLAB to a simulator to physical hardware, and find that sensing and state-estimation noise matter once the controllers leave simulation.

PROBLEM Game-theoretic controllers for resolving right of way between automated vehicles are almost always tested only in numerical simulation. So nobody knows how their yielding decisions hold up once perception, estimation noise and real actuation are involved.

METHOD They build Nash and Stackelberg decision controllers for a two-vehicle unsignalized intersection in MATLAB and Simulink. They deploy them through Quanser's real-time control software onto the Jetson AGX Orin on board each QCar 2, then test them in three stages: numerical simulation, qualitative checks in the Quanser Interactive Labs simulator, and physical scale-car runs. The scenarios cover symmetric and asymmetric approaches, leader-follower hierarchies, cases where both cars claim the same Stackelberg role, and an obstacle vehicle that does not cooperate.

RESULTS The abstract reports no numbers at all. There are no crossing times, no success or collision rates, no trial counts and no baseline comparison. The only stated finding is qualitative: the simulation and hardware results differ, and sensing and state-estimation uncertainty explain the gap.

CAVEAT This is an implementation report on small scale cars, not a new algorithm. The game solvers appear to be carried over from earlier work, and the sim-to-real gap is described, not measured. There is also no sign that any messages are exchanged. Each car seems to act on estimated states of the other, so the conflicting-role case is exactly where an explicit agreement protocol is missing, and the paper appears to observe that failure rather than fix it.

WHY IT MATTERS It sits at the edge of your conflict resolution theme. The useful part is the conflicting Stackelberg role experiment, which shows concretely what goes wrong when two vehicles hold incompatible views of who leads. That is a clear argument for negotiating roles by message, which is your focus. Nothing here speaks to message design, packet loss or partial participation. Look at the conflicting-role results and the hardware setup if you want a cheap physical testbed, and skip the rest.

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Latent Telepathy: Multi-Robot Communication with Self-Supervised Perceptual Latents

SKIM

Howard Wang, Han Zheng, Cathy Wu

Instead of sending each other positions or planned trajectories, robots broadcast the frozen perception embedding they already compute. A teammate learns from reward alone to read an occluded hazard out of it, and it does so in 99.7 percent of episodes.

PROBLEM In decentralized teams, the fact that should drive a robot's next action is often visible only to a teammate. Kinematic messages like position or trajectory cannot carry that perceptual fact. Learned MARL messages can carry it, but they are tied to one task and hard to interpret.

RESULTS With the latent message, the navigator avoids an occluded hazard in 99.7 percent of episodes. That matches a noiseless hand-designed message. Position and trajectory messages stay at chance, and a raw camera image 186 times wider is less reliable than the latent. On a physical robot, the encoder decodes the hazard in 102 of 102 live decisions. That hardware test only checks whether the hazard can be read from the latent. It is not a closed-loop multi-robot trial.

WHY IT MATTERS This is mostly peripheral to you. It is about sharing perception, the collective-perception side of V2X, not about agreeing on maneuvers. There are no proposals, acceptances, right-of-way decisions or joint trajectory optimization. Two things may still be worth a look. One is the content-controlled evaluation protocol, which you could borrow to compare intent message formats fairly. The other is their point that the decision a message informs must stay reachable by exploration, which matters if you train learned coordination policies in continuous control.

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Paying for Space: Incentive-Aware Motion Planning for Multi-Agent Collision Avoidance

SKIM

Debajyoti Chakrabarti, Anushri Dixit

They let drone operators bid for slots in a shared safe corridor, and use a VCG-style payment so that lying about your own costs never gets you a better trajectory.

PROBLEM A central airspace manager that asks operators to report their preferences, such as travel time or fuel, can be gamed, because an operator who exaggerates their costs can win better routes. Standard collision-avoidance planners assume these costs are honest, so they do nothing to stop this kind of misreporting.

RESULTS The abstract gives no numbers. It only says that simulations show safe, decentralized coordination among agents with different preferences. It names no baseline, no fleet size, no measure of how much welfare is lost, and no test of what happens when agents do misreport.

WHY IT MATTERS This sits on the edge of your area. It is about aerial vehicles, and it is a central allocation mechanism, not negotiation by exchanging messages between vehicles. It says nothing about message loss, delay or partial participation. The one idea worth taking is that right-of-way allocation can be made strategy-proof with payments. That is relevant if your conflict-resolution or auction work assumes vehicles report their urgency or costs honestly, for example in merges between vehicles from different fleets.

arXiv:2609.23256v1 PDF incremental

METHOD Each robot runs a self-supervised joint-embedding predictive encoder, in the JEPA style. The encoder is trained first, then frozen and shared by the whole team, and its latent vector is broadcast as the message. The receiver's policy learns to use that latent from task reward alone, with no decoding supervision. Their evaluation fixes bandwidth, latency, topology and receiver, and changes only what the message contains.

CAVEAT The headline comparison is set up to favor the latent. The task is built so that the deciding fact is purely perceptual, so position and trajectory messages are bound to sit at chance, and that tells us little about kinematic sharing in general. It appears to be a single occluded-hazard, one-sender-one-receiver scenario. There is no packet loss or delay sweep, no negotiation or reply, and no test with many agents or partial participation. The physical result is only 102 decoding checks, not deployed coordination.

METHOD They split planning into stages. First they build convex safe corridors through the constrained airspace. Then they allocate passage through those corridors with a mechanism modelled on VCG, where each agent pays roughly the cost its presence imposes on everyone else. That payment rule is what makes truthful reporting the best strategy, and each agent then plans its own trajectory inside the corridor it was given.

CAVEAT The VCG label hides the hard parts. Exact VCG needs an optimal allocation, and a multistage convex-corridor pipeline is almost certainly approximate, which usually breaks the truthfulness guarantee. So a reviewer would want to know whether truthfulness is actually proven for their allocation rule or just inherited from the name. The abstract also claims decentralized coordination while describing a central mechanism, and the only evidence is simulation, with no comparison to a planner that ignores incentives.

Tianhao Wu, Yiwei Lyu

An MPC controller that encodes driving rules in signal temporal logic, handles uncertain surrounding vehicles by sampling scenarios, and when the rules become impossible to satisfy, relaxes them by picking a Pareto tradeoff across several agents' objectives instead of only protecting the ego vehicle.

PROBLEM Robust STL control under uncertain, uncontrollable agents either proves that constraints hold without saying what to do when they cannot all be met, or relaxes them in a way that serves only the ego vehicle. The authors argue nobody had treated feasibility recovery as an explicit multi-objective tradeoff among agents.

RESULTS The abstract reports no numbers. It says only that the method was tested on two autonomous driving scenarios that are not named, and that it recovers feasible control with safe behaviour. It names no baselines, success rates, violation rates or computation times.

WHY IT MATTERS This is peripheral to your work. It is single-controller planning that infers what other vehicles will do, which is exactly what your work tries to replace with explicit communication, and it has no messages, proposals or acceptance steps. The one useful idea is that relaxing infeasible constraints by trading off several agents' objectives could serve as a fairness criterion when your negotiation protocol has to settle incompatible plans. But you would have to build the communication layer yourself.

METHOD They use scenario-based MPC, meaning the other agents' stochastic behaviour is represented by sampled trajectories and the STL constraints are enforced across those samples. When the problem becomes infeasible, the constraint relaxation is posed as a Pareto optimization over the objectives of several agents, so the controller picks a compromise rather than maximizing ego utility. Scenario-optimization sampling bounds then give a probabilistic certificate on how often the STL specification is violated.

CAVEAT The other agents are uncontrollable and do not communicate, so the Pareto tradeoff is decided by one controller using guessed preferences for the others, not agreed through negotiation. A reviewer would also ask how to pick a single point from the Pareto front in real time, whether the violation certificate still holds once constraints are relaxed, and whether two hand-built scenarios with no stated baselines can show anything beyond a proof of concept.

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